

# FPROS6-A1700-OSB

Serie FPROS • R513A • senza serbatoio / without tank



FPROS6-A1700-OSB

RESA FRIGORIFERA • COOLING CAPACITY

**1.113** kW

EER • 12/7 °C • AIR 35 °C

**3,00**

RESA NOMINALE • 15 °C • AIR 25 °C

**1.699** kW

PORTATA ACQUA • WATER FLOW

**191,5** m<sup>3</sup>/h

|                            |                 |                                        |                        |
|----------------------------|-----------------|----------------------------------------|------------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 13.600 × 2.050 × 2.400 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                      |
| Ventilatori / Fans         | 24              | Taglia / Size                          | 1700                   |
| Free cooling (ΔT 8 K)      | 1.200 kW *      | Campo aria esterna / Ambient air range | -5 / +47 °C            |
| Refrigerante / Refrigerant | R513A • GWP 631 |                                        |                        |

## Resa in sintesi / Capacity at a glance kW

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 1.246 | 1.180 | 1.113 | 1.046 | 977   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.699 | 1.614 | 1.526 | 1.438 | 1.348 |

Refrigeratori con free cooling integrato e compressori a vite, per grandi potenze che lavorano tutto l'anno.

Chillers with integrated free cooling and screw compressors, for large capacities running all year round.

# Dati tecnici

## Technical data

FPROS6 - A1700 - OSB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.699,31 kW              |
| Resa frigorifera / Cooling capacity | 1.461.407 kcal/h         |
| Potenza assorbita / Power input     | 316,88 kW                |
| EER / EER                           | 5,36                     |
| Portata acqua / Water flow          | 292,28 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.113,35 kW              |
| Potenza assorbita / Power input     | 371,17 kW                |
| EER / EER                           | 3,00                     |
| ESEER / ESEER                       | 3,96                     |
| Portata acqua / Water flow          | 191,50 m <sup>3</sup> /h |

### Componenti / Components

|                                               |                      |
|-----------------------------------------------|----------------------|
| Compressori / Compressors                     | 2 × Vite             |
| Circuiti frigoriferi / Refrigerant circuits   | 2                    |
| Ventilatori / Fans                            | 24                   |
| Free cooling (ΔT 8 K) / Free cooling (8 K ΔT) | 1.200 kW *           |
| Ventilatori free cooler / Free cooler fans    | 26                   |
| Condensatore / Condenser                      | a microcanali        |
| Evaporatore / Evaporator                      | a fascio tubiero     |
| Refrigerante / Refrigerant                    | R513A · GWP 631 · A1 |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 13.600 × 2.050 × 2.400 mm |
| Campo aria esterna / Ambient air range                                                    | -5 / +47 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

\* Potenza di free cooling indicativa, da confermare in fase di offerta. / Indicative free cooling capacity, to be confirmed at quotation stage.

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 · 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 1.091,1 kW   |
| MEG 30 % | × 0,950 | 1.057,7 kW   |
| MEG 40 % | × 0,920 | 1.024,3 kW   |
| MPG 20 % | × 0,975 | 1.085,5 kW   |
| MPG 30 % | × 0,960 | 1.068,8 kW   |
| MPG 40 % | × 0,950 | 1.057,7 kW   |

# Prestazioni Performance

FPROS6-A1700-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35    | 40    | 45    |
|---------------|-------|-------|-------|-------|-------|-------|-------|
| 1             | 1.074 | 1.024 | 972   | 919   | 865   | 810   | 755   |
| 2             | 1.121 | 1.069 | 1.015 | 960   | 904   | 847   | 789   |
| 3             | 1.167 | 1.113 | 1.058 | 1.001 | 942   | 884   | 824   |
| 4             | 1.217 | 1.161 | 1.103 | 1.044 | 984   | 923   | 861   |
| 5             | 1.266 | 1.208 | 1.149 | 1.088 | 1.025 | 962   | 898   |
| 6             | 1.319 | 1.259 | 1.197 | 1.134 | 1.069 | 1.004 | 938   |
| 7             | 1.372 | 1.310 | 1.246 | 1.180 | 1.113 | 1.046 | 977   |
| 8             | 1.428 | 1.364 | 1.298 | 1.230 | 1.160 | 1.091 | 1.019 |
| 9             | 1.484 | 1.418 | 1.349 | 1.279 | 1.207 | 1.135 | 1.061 |
| 10            | 1.544 | 1.475 | 1.404 | 1.332 | 1.257 | 1.183 | 1.106 |
| 11            | 1.603 | 1.532 | 1.459 | 1.384 | 1.307 | 1.230 | 1.151 |
| 12            | 1.667 | 1.593 | 1.517 | 1.440 | 1.360 | 1.280 | 1.199 |
| 13            | 1.730 | 1.654 | 1.576 | 1.496 | 1.413 | 1.331 | 1.247 |
| 14            | 1.797 | 1.718 | 1.638 | 1.555 | 1.470 | 1.384 | 1.298 |
| 15            | 1.864 | 1.783 | 1.699 | 1.614 | 1.526 | 1.438 | 1.348 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| 1             | 254 | 278 | 304 | 333 | 365 | 400 | 439 |
| 2             | 255 | 279 | 305 | 334 | 366 | 401 | 440 |
| 3             | 256 | 280 | 306 | 335 | 367 | 403 | 442 |
| 4             | 257 | 281 | 307 | 336 | 368 | 404 | 443 |
| 5             | 258 | 282 | 308 | 337 | 369 | 405 | 444 |
| 6             | 259 | 282 | 309 | 338 | 370 | 406 | 446 |
| 7             | 260 | 283 | 309 | 339 | 371 | 407 | 447 |
| 8             | 261 | 284 | 310 | 339 | 372 | 408 | 448 |
| 9             | 262 | 285 | 311 | 340 | 373 | 409 | 449 |
| 10            | 263 | 286 | 312 | 341 | 374 | 410 | 450 |
| 11            | 264 | 287 | 313 | 342 | 375 | 411 | 452 |
| 12            | 265 | 288 | 314 | 343 | 376 | 412 | 453 |
| 13            | 266 | 289 | 315 | 344 | 377 | 413 | 454 |
| 14            | 267 | 290 | 316 | 345 | 378 | 414 | 455 |
| 15            | 268 | 291 | 317 | 346 | 379 | 415 | 456 |

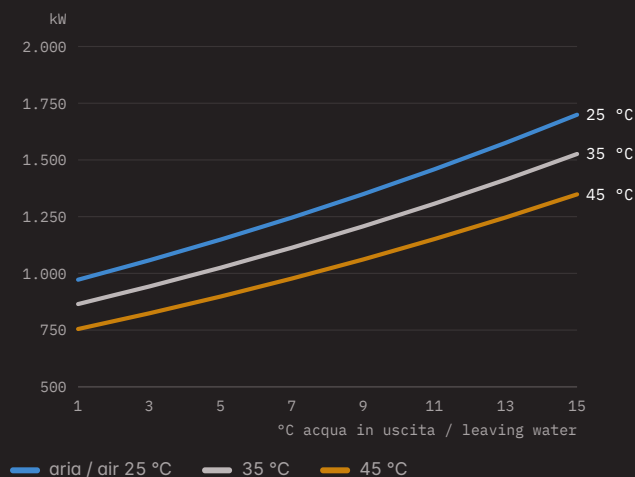
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   |
|---------------|------|------|------|------|------|------|------|
| 1             | 4,22 | 3,68 | 3,20 | 2,76 | 2,37 | 2,03 | 1,72 |
| 2             | 4,39 | 3,83 | 3,33 | 2,88 | 2,47 | 2,11 | 1,79 |
| 3             | 4,56 | 3,98 | 3,46 | 2,99 | 2,57 | 2,19 | 1,87 |
| 4             | 4,73 | 4,14 | 3,60 | 3,11 | 2,67 | 2,29 | 1,94 |
| 5             | 4,91 | 4,29 | 3,73 | 3,23 | 2,78 | 2,38 | 2,02 |
| 6             | 5,10 | 4,46 | 3,88 | 3,36 | 2,89 | 2,47 | 2,10 |
| 7             | 5,28 | 4,62 | 4,03 | 3,49 | 3,00 | 2,57 | 2,19 |
| 8             | 5,48 | 4,80 | 4,18 | 3,62 | 3,12 | 2,67 | 2,27 |
| 9             | 5,67 | 4,97 | 4,34 | 3,76 | 3,24 | 2,77 | 2,36 |
| 10            | 5,88 | 5,16 | 4,50 | 3,90 | 3,36 | 2,88 | 2,46 |
| 11            | 6,08 | 5,34 | 4,66 | 4,04 | 3,49 | 2,99 | 2,55 |
| 12            | 6,30 | 5,53 | 4,83 | 4,20 | 3,62 | 3,11 | 2,65 |
| 13            | 6,51 | 5,73 | 5,00 | 4,35 | 3,75 | 3,22 | 2,75 |
| 14            | 6,74 | 5,93 | 5,18 | 4,51 | 3,89 | 3,34 | 2,85 |
| 15            | 6,96 | 6,13 | 5,36 | 4,67 | 4,03 | 3,46 | 2,96 |

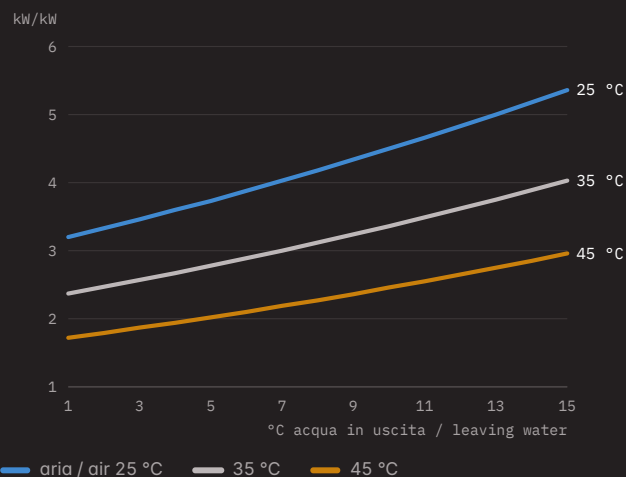
# Curve e dimensioni

## Curves and dimensions

### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 13.600 × 2.050 × 2.400 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

FPROS6-A1700-OSB · 55 optional disponibili per questa configurazione · 55 options available for this configuration

| Pompe / Pumps                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pompa 195 m³/h – prevalenza 25 m<br>Pump 195 m³/h – head 25 m                                                                                                                              |
| Pompa 195 m³/h – prevalenza 40 m<br>Pump 195 m³/h – head 40 m                                                                                                                              |
| Pompa doppia montata 195 m³/h – prevalenza 25 m<br>Twin pump, factory -fitted, 195 m³/h – head 25 m                                                                                        |
| Pompa doppia montata 195 m³/h – prevalenza 40 m<br>Twin pump, factory -fitted, 195 m³/h – head 40 m                                                                                        |
| Serbatoi e vasi di espansione / Tanks and expansion vessels                                                                                                                                |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin AISI 304 pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin AISI 304 pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin pump                                              |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin pump                                              |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory-fitted single pump |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump |
| Vaso di espansione 100 l<br>Expansion vessel 100 l                                                                                                                                         |
| Vaso di espansione 200 l<br>Expansion vessel 200 l                                                                                                                                         |
| Vaso di espansione 300 l<br>Expansion vessel 300 l                                                                                                                                         |
| Vaso di espansione 500 l<br>Expansion vessel 500 l                                                                                                                                         |
| Vaso di espansione 750 l<br>Expansion vessel 750 l                                                                                                                                         |

| Valvole e filtri / Valves and filters                    |
|----------------------------------------------------------|
| Elettrovalvola 2"<br>Solenoid valve 2"                   |
| Elettrovalvola 3"<br>Solenoid valve 3"                   |
| Elettrovalvola 4"<br>Solenoid valve 4"                   |
| Elettrovalvola 5"<br>Solenoid valve 5"                   |
| Elettrovalvola 6"<br>Solenoid valve 6"                   |
| Filtro a dischi in acciaio 2"<br>Steel disc filter 2"    |
| Filtro a dischi in acciaio 3"<br>Steel disc filter 3"    |
| Filtro a dischi in acciaio 4"<br>Steel disc filter 4"    |
| Filtro meccanico a Y 2"<br>Y-strainer 2"                 |
| Filtro meccanico a Y 3"<br>Y-strainer 3"                 |
| Filtro meccanico a Y 4"<br>Y-strainer 4"                 |
| Filtro meccanico a Y 5"<br>Y-strainer 5"                 |
| Filtro meccanico a Y 6"<br>Y-strainer 6"                 |
| Valvola a 3 vie 2" miscelatrice<br>3-way mixing valve 2" |
| Valvola a 3 vie 3" miscelatrice<br>3-way mixing valve 3" |
| Valvola a 3 vie 4" miscelatrice<br>3-way mixing valve 4" |
| Valvola a 3 vie 5" miscelatrice<br>3-way mixing valve 5" |
| Valvola a 3 vie 6" miscelatrice<br>3-way mixing valve 6" |
| Valvola a 3 vie 2" on/off<br>3-way valve 2" on/off       |
| Valvola a 3 vie 3" on/off<br>3-way valve 3" on/off       |
| Valvola a 3 vie 4" on/off<br>3-way valve 4" on/off       |
| Valvola a 3 vie 5" on/off<br>3-way valve 5" on/off       |

# Optional — 2/2

## Options

FPROS6-A1700-OSB · 55 optional disponibili per questa configurazione · 55 options available for this configuration

### Valvole e filtri / Valves and filters

Valvola a 3 vie 6" on/off  
3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori  
Compressor acoustic enclosure

Compressore inverter  
Inverter compressor

Desurriscaldatore a piastre  
Plate desuperheater

Rilevatore fughe gas  
Refrigerant leak detector

### Ventilazione e clima / Fans and climate

Inverter ventilatori  
Fan inverter

Ventilatori EC  
EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)  
Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)  
RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)  
RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete  
Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C  
Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla  
Spring anti-vibration mounts

Antivibranti in gomma  
Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
  - Free cooling: potenza riferita a una differenza di 8 K tra acqua e aria esterna; valore indicativo, da confermare in fase di offerta.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.
  - Free cooling: capacity referred to an 8 K difference between water and outdoor air; indicative value, to be confirmed at quotation stage.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

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Continuity by design